Practice Exam - 1

Question 113 of 117

A 66-year-old woman with a history of mild COPD and heart failure with reduced ejection fraction (EF 25%) experiences reduced exertional tolerance. She was recently hospitalized and treated for both a COPD flare and a heart failure exacerbation. In clinic, vital signs are BP 105/72 mm Hg, heart rate 90 bpm, SpO2 96% on room air, BMI 31 kg/m2. She is referred for a cardiopulmonary exercise test.

Results are as follows:

Exercise time: 11 minutes and 9 seconds (stops due to dyspnea) Peak workload: 125 Watts (4.3 METS) Absolute peak VO2: 1371 ml/min

Peak heart rate: 144 bpm
Normalized peak VO2: 14.1 ml/Kg/min (57% predicted) Maximum ventilatory volume: 150 L/min
Peak ventilation: 99 L/min

Respiratory exchange ratio: 1.39
VE/VCO2: 32
SpO2 at peak activity: 92%

What is the primary cause of her exertional limitation?

A. Cardiac limitation

B. Deconditioning

C. Cannot interpret due to obesity

D. Pulmonary limitation

Commentary References

Testing point: The Board-certified AHFTC specialist should be able identify the cause of exertional limitation on cardiopulmonary exercise test.

Answer: A. Cardiac limitation.

The patient has a moderate reduction in exertional capacity, having only reached 57% of her predicted peak VO2. VE/VCO2 is elevated as well but this does not always help distinguish between cardiac and pulmonary limitation, as abnormalities in either can result in ventilatory inefficiency. The patient's estimated oxygen pulse, however, is < 10 ml/min (1371/144) which argues for reduced exercise stroke volume as the culprit for the patient's limitation.

Answer D: Although the patient's SpO2 drops to 92% at peak activity, the patient retains a breathing reserve of 0.34 (1 - 99/150) arguing against a pulmonary limitation.

Answer C: The patient's BMI is indeed elevated but not enough to warrant discounting the peak VO2.

Answer B: The patient reaches an adequate respiratory exchange ratio, which indicates achievement of maximal volitional effort and no limitation referable to deconditioning.